

Angular and Rotation-Rate Scaling of Speckle Intensity Autocorrelations from a Rotating Diffuser

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Abstract

We measure normalized intensity autocorrelations of laser light scattered from a rotating ground-glass diffuser using a single-photon counting module and an FPGA-based logarithmic multi- τ correlator. Traces are recorded at scattering angles of approximately 0° , 10° , 15° , and 20° for several rotation periods. On a logarithmic delay axis the decays are well described by a four-parameter complementary-error-function (erfc) sigmoid in $\ln \tau$, from which we extract a midpoint time τ_{50} . At fixed angle, $1/\tau_{50}$ increases linearly with angular speed ω , consistent with the prediction that diffuser angular displacement $\omega\tau$ —not absolute time—governs decorrelation. The slope of that relation grows approximately in proportion to $\sin \theta$ for the well-aligned off-axis angles, in qualitative agreement with far-field models of rigidly rotating speckle. A nominal forward-scattering run still shows finite decay and does not follow the same $\sin \theta$ line, which we attribute to imperfect on-axis alignment. We summarize what is established by these data, how it compares with classical rotating-diffuser literature, and what measurements would most usefully extend the result.

1 Introduction

When coherent light illuminates a rough surface, the scattered field forms a speckle pattern whose spatial and temporal statistics encode the surface structure and its motion [1, 2]. For an in-plane rotating diffuser, the pattern evolves as new regions of the rough surface enter the illuminated spot and as the far-field interference pattern rotates or translates. Intensity fluctuations at a fixed detector therefore decorrelate on a time scale set by the rotation rate and by the optical geometry [3–6].

This physics is both a classical metrology problem—speckle correlation and zero-crossing methods have long been used to infer surface velocity [3, 7]—and a practical tool. Rotating or translating diffusers are routinely used to reduce spatial coherence, homogenize beams, and synthesize pseudo-thermal light [1]. More recently, the same spatiotemporal intensity fields have been discussed as controllable random optical drives for thermal-transport and system-identification experiments. In all of those settings one needs a quantitative map from rotation rate and observation angle to the temporal spectrum of the intensity.

The present experiment is a focused test of the simplest scaling predictions. If the diffuser presents a fixed sequence of scattering configurations as it rotates, then changing the angular speed ω should only change the *playback rate* of that sequence. At fixed scattering angle θ , normalized correlation curves should then collapse when plotted against the angular displacement $\omega\tau$, which

implies

$$\tau_c(\theta, \omega) = \frac{\Delta\theta_c(\theta)}{\omega}, \quad \text{or equivalently} \quad \frac{1}{\tau_c} \propto \omega. \quad (1)$$

Here $\Delta\theta_c$ is a characteristic diffuser rotation angle required to decorrelate the detected intensity. Idealized far-field models of *centered*, rigidly rotating circular illumination further predict that the correlation time shortens with angle roughly as $1/\sin\theta$ [4, 6], so that the slope in Eq. (1) itself should increase with $\sin\theta$.

Our goals in this draft are deliberately limited. We (i) acquire multi- τ photon autocorrelations over a grid of angles and rotation periods, (ii) fit each curve with a log-delay sigmoid matched to the multi- τ sampling, (iii) test Eq. (1), and (iv) compare the extracted slopes with a $\sin\theta$ trend. We do *not* yet claim a full reconstruction of the sample-plane spatiotemporal spectrum, nor a precision measurement of illuminated diameter.

2 Background

2.1 Intensity autocorrelation

Let $I(t)$ be the photocount-derived intensity at a fixed detector. The normalized second-order correlation commonly used in dynamic light scattering [8, 9] is

$$g^{(2)}(\tau) = \frac{\langle I(t) I(t + \tau) \rangle}{\langle I(t) \rangle^2}. \quad (2)$$

For a stationary process, $g^{(2)}(\tau) \rightarrow 1$ at long delay. The intercept $g^{(2)}(0) - 1$ is the measured speckle contrast β ; it is reduced below the ideal single-mode value of unity by finite detector area, polarization mixing, and temporal averaging [1].

2.2 Rotating-diffuser expectations

Saleh, Tivolacci, and Soper demonstrated velocity measurement from rotating-object speckle correlations [3]. Churnside analyzed speckle from a rotating diffuse object in detail [4]. Yoshimura and coworkers distinguished rotational motion of the pattern from speckle “boiling” [5]. Yura, Rose, and Hanson gave ABCD-system expressions for in-plane rotation [6]. In the idealized limit of a circular pupil centered on the rotation axis, the short-delay field correlation takes a jinc form and the intensity correlation follows from the Siegert relation, yielding

$$g^{(2)}(\tau) - 1 \approx \beta \left[\frac{2J_1(X)}{X} \right]^2, \quad X \propto D \omega \sin\theta \tau / \lambda, \quad (3)$$

with illuminated diameter D and wavelength λ . Equation (3) implies both the ω^{-1} scaling of Eq. (1) and an angle factor $\sin\theta$. Off-axis illumination, beam nonuniformity, and surface boiling replace or supplement rigid rotation with additional decorrelation channels [5, 6].

2.3 Why an $\text{erfc}(\ln \tau)$ fit?

Our correlator returns lags on an approximately logarithmic grid. On a $\log \tau$ axis the measured decays look like smooth sigmoids rather than pure exponentials. A convenient phenomenological model, already used in our laboratory for multi- τ dynamic-light-scattering traces, is

$$g^{(2)}(\tau) = A \text{erfc}\left(\frac{\ln \tau - B}{C_w}\right) + D, \quad (4)$$

where erfc is the complementary error function. The midpoint

$$\tau_{50} = e^B \quad (5)$$

is a robust characteristic time on the log-delay axis. The parameter C_w sets the transition width in units of $\ln \tau$, and the step height $2A$ estimates the contrast β when the baseline $D \approx 1$. This form does not replace a first-principles jinc model; it is a data-reduction tool well matched to log-spaced sampling. Pilot comparisons on representative runs showed that a single exponential ($\alpha = 1$ stretched exponential) is a poor description of these diffuser curves, whereas Eq. (4) and free-shape stretched exponentials both perform well. Ideal jinc² curves alone leave systematic Airy-lobe residuals and are not used for the scaling analysis below.

3 Experiment

3.1 Apparatus

A diode laser (operating near 650 nm, drive 3.31 V/0.033 A) was focused onto a rotating ground-glass diffuser mounted at the center of a diffractometer-style rotation stage. The diffuser grit was not recorded for this run and is left unspecified. Scattered light was collected into a Thorlabs single-photon counting module with a 20 μm active diameter. An iris kept the mean count rate of order 1 MHz (measured rates in the present files span roughly 1 MHz to 2.3 MHz). Collection optics beyond the iris were not fully characterized in this first survey.

Photon arrival times were processed by a custom FPGA logarithmic multi- τ binary autocorrelator clocked at $f = 12$ MHz. Host software (`simple_multitau.m`, a single-angle simplification of a multi-angle acquisition code) requested 10 s acquisitions, read the correlation bins over SPI, and saved Octave text exports together with a semilog- x preview figure.

3.2 Run matrix

Autocorrelations were taken at four nominal scattering angles—20°, 15°, 10°, and 0°—with six rotation periods at each angle (two periods share the label `test027` at 15°). Early unlabeled filenames correspond to 20°. Rotation periods T_{rot} were timed with a stopwatch for one revolution and encoded in the filenames; angular speed is taken as

$$\omega = \frac{2\pi}{T_{\text{rot}}}. \quad (6)$$

The nominal 0° setting required a very small iris for on-axis count-rate control and was only roughly aligned; we treat those runs as a poorly constrained *small angle*, not as true forward scattering.

3.3 Normalization

The correlator returns coincidence counts divided by the total photon count. With clock duration T_{clk} (in cycles) and total counts N , the mean occupancy per clock cycle is $\langle n \rangle = N/T_{\text{clk}}$. We form

$$g^{(2)}(\tau) = \frac{\text{AC}(\tau)}{\langle n \rangle} = \text{AC}(\tau) \frac{T_{\text{clk}}}{N}, \quad (7)$$

so that the long-delay baseline approaches unity. The first ten lags ($\tau \lesssim 0.8 \mu\text{s}$) show a sharp afterpulsing rise and are excluded from all fits.

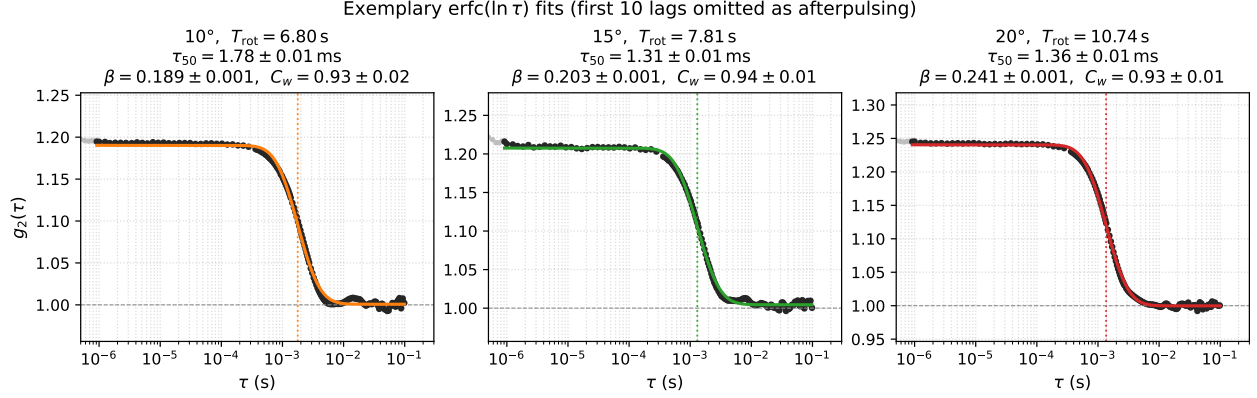


Figure 1: Exemplary normalized autocorrelations and $\text{erfc}(\ln \tau)$ fits at three scattering angles. Grey early points are omitted from the fit (afterpulsing). The dotted vertical line marks τ_{50} ; the shaded band is the bootstrap uncertainty on τ_{50} .

4 Data analysis

Each trace was fitted to Eq. (4) over $0.8 \mu\text{s} \lesssim \tau \leq 0.1 \text{ s}$ by nonlinear least squares. Parameter uncertainties were estimated by residual bootstrap (400 resamples), which is more conservative than formal covariance errors when residuals contain smooth systematic structure. From each fit we retain τ_{50} , $\beta \approx 2A$, C_w , and the baseline D .

At each fixed angle we then fit

$$\frac{1}{\tau_{50}} = m\omega + b \quad (8)$$

with a floated intercept b . Ideal playback-rate scaling predicts $b \approx 0$ and a slope $m = 1/\Delta\theta_c$. Finally we plot the floated slopes $m(\theta)$ against $\sin \theta$ and compare them with a line through the origin fitted to the 10° , 15° , and 20° points only.

5 Results

5.1 Shape of individual correlations

Figure 1 shows three representative fits at mid-range rotation periods for 10° , 15° , and 20° . In each case the $\text{erfc}(\ln \tau)$ model captures the plateau, the sigmoidal shoulder on the log-delay axis, and the return to baseline. Extracted midpoints are of order 0.7 ms to 2 ms for these examples, with bootstrap uncertainties at the percent level or better. Contrast β increases from about 0.19 at 10° to about 0.24 at 20° in these exemplars, consistent with the broader data set.

5.2 Scaling with rotation rate

Figure 2 plots $1/\tau_{50}$ versus ω for all angles, with a separate floated-intercept line at each angle. The off-axis families (10° – 20°) are strikingly linear. Intercepts are small compared with the span of the data, so the leading behavior is the through-origin relation predicted by Eq. (1). Faster rotation (larger ω) systematically yields faster decorrelation (larger $1/\tau_{50}$), in quantitative proportion at fixed angle.

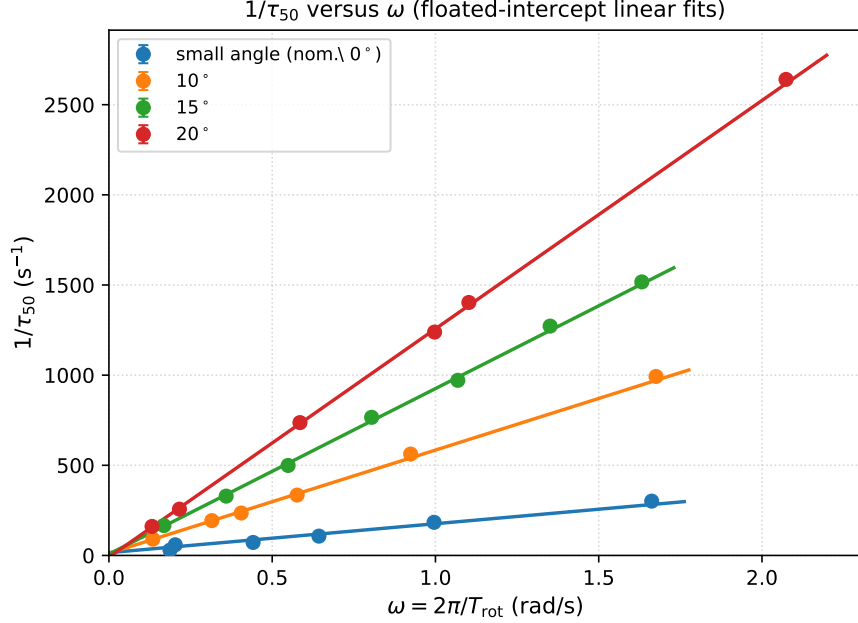


Figure 2: Inverse midpoint time versus angular speed. Solid lines are weighted linear fits with floated intercepts. Error bars are smaller than the markers in most cases.

Table 1: Floated-intercept slopes of $1/\tau_{50} = m\omega + b$. Uncertainties are from the weighted line fit using bootstrap errors on τ_{50} .

Angle	m	$u(m)$	b (s ⁻¹)	$u(b)$
small (nom. 0°)	160.6	1.9	15.4	0.7
10°	572.6	3.0	11.9	0.9
15°	917.8	3.8	7.7	1.7
20°	1266.6	3.8	-9.7	1.2

Table 1 lists the floated slopes. Equivalent angular decorrelation scales $\Delta\theta_c \equiv 1/m$ (using the origin-forced slope from the companion analysis) shrink from roughly 0.10° at 10° to 0.05° at 20° : less diffuser rotation is needed to refresh the speckle at larger scattering angle.

5.3 Scaling with angle

Figure 3 shows the floated slopes versus $\sin\theta$. The 10° , 15° , and 20° points fall close to a single line through the origin,

$$m(\theta) \approx k \sin\theta, \quad k = (3.57 \pm 0.11) \times 10^3, \quad (9)$$

fitted while excluding the nominal 0° point. That trend is exactly the angle factor expected from Eq. (3) if τ_{50} tracks the same combination $\omega \sin\theta$ that sets the jinc argument—even though the detailed jinc *shape* is not required for the scaling test.

The small-angle point lies well above the $\sin\theta = 0$ prediction of zero slope. Finite decorrelation on axis is expected if the detector is not at true forward scattering, if the beam is not centered on the rotation axis, or if boiling/wobble contribute angle-independent dynamics [5]. Given the

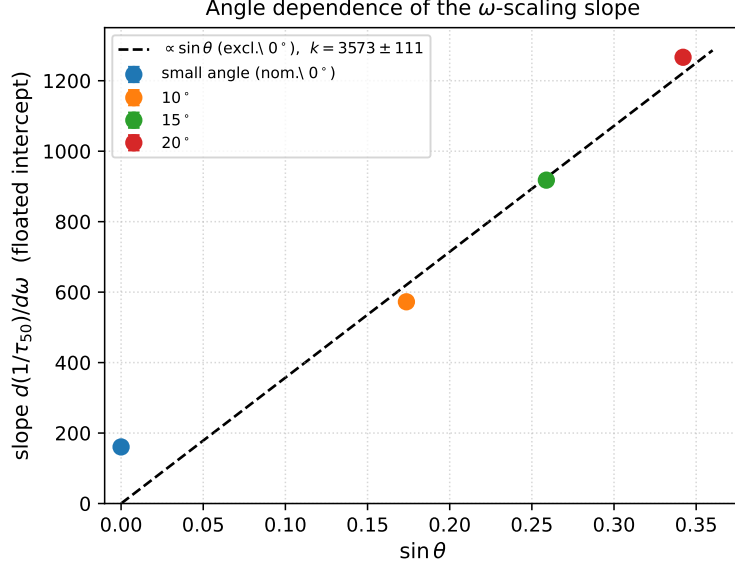


Figure 3: Floated slopes from Fig. 2 versus $\sin \theta$. The dashed line is a through-origin fit to the 10°, 15°, and 20° points only; the nominal 0° point is shown but excluded from that fit.

documented rough alignment of the 0° runs, we interpret that point as a diagnostic of nonideal geometry rather than a failure of the off-axis scaling law.

6 Comparison with theory and prior work

Agreement. The data affirm the playback-rate hypothesis of Eq. (1) at fixed angle and the $\sin \theta$ growth of the rate constant for 10°–20°. Both are standard expectations from rotating-diffuser speckle theory [4, 6] and from earlier correlation-based velocimetry [3, 7]. The practical advance here is a compact, photon-counting multi- τ survey that makes the two-parameter scaling (ω, θ) visible in a single afternoon’s run matrix, with modern log-lag instrumentation and a fit function matched to that lag grid.

Limits of the present comparison. We have not yet determined the illuminated diameter D independently, so we do not convert the measured k into a precision test of the prefactor in Eq. (3). Collection optics are incompletely specified, so absolute contrast β should not be over-interpreted. The $\text{erfc}(\ln \tau)$ midpoint τ_{50} is not identical to the $1/e$ time of a jinc² or stretched exponential; it is a consistent empirical clock for scaling tests, not a claim that the microscopic correlation equals an error function.

What is new relative to classical papers. Classical studies established the mechanisms and often emphasized velocity extraction or imaging-geometry taxonomy [3–5]. The present draft’s contribution is narrower but concrete: a clear experimental collapse of $\tau_{50}(\omega)$ at several angles on a photon-counting FPGA correlator, plus a direct $\sin \theta$ test of the resulting slopes, including an honest treatment of a misaligned near-forward channel.

7 Conclusions and outlook

Within the off-axis data set, prediction and measurement agree on two central points: (1) decorrelation rates scale linearly with rotation speed at fixed angle, and (2) those rates increase approximately proportionally to $\sin \theta$. A phenomenological $\text{erfc}(\ln \tau)$ fit provides stable τ_{50} values for that test. The near-forward runs remind us that geometry must be controlled before interpreting small-angle residuals as intrinsic boiling.

The most impactful next experiments are, in our view:

1. **Collapse of full curves**, not only τ_{50} : plot contrast-normalized $g^{(2)}$ against $\omega\tau$ (and against $\omega\tau \sin \theta$) to test shape invariance [6].
2. **Revolution echoes**: extend the lag window over one or more full periods to measure recurrence amplitude—a direct probe of rigid versus boiling dynamics [5].
3. **Controlled beam offset R** : vary the illuminated radius from the axis to separate tangential-displacement scaling $\ell_c \sim R\omega\tau_c$ from pure angular scaling.
4. **Camera measurement of $S_I(\mathbf{k}, \omega)$** at the intended sample/heating plane, which single-point autocorrelations cannot fully determine.
5. **Documented grit, spot size, and collection NA**, enabling a quantitative jinc or ABCD comparison rather than slope ratios alone.

Items 1–3 stay within the present photon-correlator setup and would most directly tighten the link between this data set and the classical theory. Items 4–5 open the path from characterization of the diffuser drive to its use as a calibrated stochastic optical field.

Acknowledgments

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